

Topic Covered:

- Discrete Probability Distributions

1. Determine whether the distribution represents a probability distribution. If it does not, state why.

a.

X	3	6	8	12
P(X)	0.3	0.5	0.7	-0.8

b.

X	-2	0	2	5
P(X)	0.3	0.4	0.2	0.1

c.

X	1	2	3	4
P(X)	0.4	0.3	0.2	1

Answer:

- a) No; probabilities cannot be negative (-0.8) and the sum of the probabilities is not 1.
 b) Yes
 c) No; the sum of the probabilities is greater than 1.

2. Probability Distribution: A bag contains five balls numbered 1, 2, 4, 7, and *. Choose two balls at random without replacement and add the numbers. If one ball has the *, double the amount on the other ball. Construct the probability distribution for this random variable X and calculate μ , σ^2 and σ .

Answer:

The set of balls is {1, 2, 4, 7, *}. We list all 20 ordered pairs (permutations) and calculate the value of the random variable X for each. The rules for X are: Sum if no * is drawn, and Double the other ball's value if one * is drawn.

Pair	X Value	Pair	X Value	Pair	X Value	Pair	X Value	Pair	X Value
(1, 2)	3	(2, 1)	3	(4, 1)	5	(7, 1)	8	(*, 1)	2
(1, 4)	5	(2, 4)	6	(4, 2)	6	(7, 2)	9	(*, 2)	4
(1, 7)	8	(2, 7)	9	(4, 7)	11	(7, 4)	11	(*, 4)	8
(1, *)	2	(2, *)	4	(4, *)	8	(7, *)	14	(*, 7)	14

We count the frequency (f) of each unique X value. The probability $P(X = x)$ is $\frac{f}{20}$.

x	2	3	4	5	6	8	9	11	14
P(X = x)	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$



The Mean (expected value) is calculated as $\mu = E[X] = \sum x \cdot P(X = x)$:

$$\mu = 0.2 + 0.3 + 0.4 + 0.5 + 0.6 + 1.6 + 0.9 + 1.1 + 1.4 = 7.$$

The variance is calculated using the formula $\sigma^2 = E[X^2] - \mu^2$, where $E[X^2] = \sum x^2 \cdot P(X = x)$.

$$E[X^2] = 2^2(0.1) + 3^2(0.1) + 4^2(0.1) + 5^2(0.1) + 6^2(0.1) + 8^2(0.2) + 9^2(0.1) + 11^2(0.1) + 14^2(0.1).$$

$$\sigma^2 = E[X^2] - \mu^2 = 61.6 - (7)^2 = 12.6.$$

The standard deviation is the positive square root of the variance: $\sigma = \sqrt{\sigma^2}$.

$$\sigma = \sqrt{12.6} \approx 3.55.$$

3. People Who Have Some College Education: Fifty three percent of all persons in the U.S. population have at least some college education. Choose 10 persons at random. Find the probability that
- Exactly one-half have some college education
 - At least 5 do not have any college education
 - Fewer than 5 have some college education

Answer:

The probability of getting exactly k successes in n trials is given by the binomial probability formula:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

$$\text{a. Exactly one-half have ... : } P(X = 5) = \binom{10}{5} (0.53)^5 (0.47)^{10-5} \approx 0.2423.$$

$$\text{b. At least 5 do not have ... : } P(Y \geq 5) = \sum_{k=5}^{10} \binom{10}{k} (0.47)^k (0.53)^{10-k} \approx 0.548.$$

$$\text{c. Fewer than 5 have ... : } P(X < 5) = \sum_{k=0}^4 \binom{10}{k} (0.53)^k (0.47)^{10-k} \approx 0.306.$$

4. Survey of High School Seniors: Of graduating high school seniors, 14% said that their generation will be remembered for their social concerns. If 7 graduating seniors are selected at random, find the probability that either 2 or 3 will agree with that statement.

Answer:

The probability that either 2 or 3 graduating seniors will agree with the statement is $P(X = 2 \text{ or } X = 3)$.

$$P(X = 2 \text{ or } X = 3) = P(X = 2) + P(X = 3)$$

$$= \binom{7}{2} (0.14)^2 (0.86)^{7-2} + \binom{7}{3} (0.14)^3 (0.86)^{7-3} \approx 0.1936 + 0.0526 \approx 0.2462.$$

5. Colors of Flowers A nursery provides red impatiens for commercial landscaping. If 5% are variegated instead of pure red, find the probability that in an order for 200 plants, exactly 14 are variegated.

Answer:

The probability of finding exactly x successes is given by the Poisson formula:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \lambda = n \cdot p$$

For the Poisson approximation to the binomial distribution, the mean (λ) is calculated as the product of the number of trials (n) and the probability of success (p).

We want to find $P(X = 14)$ with $\lambda = (200)(0.05) = 10$:

$$P(X = 14) = \frac{e^{-10} (10)^{14}}{14!} \approx 0.05208$$

